Duality III: Marginal Costs and Dual Simplex

How much does a changed requirement cost? What can we do when the old basis no longer gives feasible quantities?

First interpret optimal dual variables as marginal costs. Then use dual simplex to restore primal feasibility while keeping a valid dual bound.

- Marginal costs
- Dual-simplex pivots
- Primal and dual geometry
- Complete proofs and further exploration

Begin with a primal solution and its matching dual bound

Primal · Example 4.6

$$\begin{aligned} \min \quad & 13x_1 + 10x_2 + 6x_3 \\ & 5x_1 + x_2 + 3x_3 = 8, \\ & 3x_1 + x_2 = 3, \qquad x \geq 0. \end{aligned}$$

Dual

$$\begin{aligned} \max \quad & 8p_1 + 3p_2 \\ & 5p_1 + 3p_2 \leq 13, \\ & p_1 + p_2 \leq 10, \quad 3p_1 \leq 6, \\ & p_1, p_2 \in \mathbb{R}. \end{aligned}$$

Each p_i corresponds to primal equality i. Equality rows make p unrestricted.

$$x^* = (1, 0, 1), \quad 13(1) + 6(1) = 19; \qquad p^* = (2, 1), \quad 8(2) + 3(1) = 19.$$

Both points are feasible and their values match, so both are optimal. We now change one requirement at a time.

Change requirement 1 from 8 to 9

Primal · first target: 8 → 9

$$\begin{aligned} \min \quad & 13x_1 + 10x_2 + 6x_3 \\ & 5x_1 + x_2 + 3x_3 = 9, \\ & 3x_1 + x_2 = 3, \qquad x \geq 0. \end{aligned}$$

Dual

$$\begin{aligned} \max \quad & 9p_1 + 3p_2 \\ & 5p_1 + 3p_2 \leq 13, \\ & p_1 + p_2 \leq 10, \quad 3p_1 \leq 6, \\ & p_1, p_2 \in \mathbb{R}. \end{aligned}$$

Keep $x_2 = 0$. Then $3x_1 = 3$, and $5x_1 + 3x_3 = 9$, giving $x = (1, 0, 4/3)$.

$$13(1) + 6(4/3) = 21, \qquad 9(2) + 3(1) = 21.$$

The dual point $p = (2, 1)$ remains feasible: only its objective coefficients changed, not its constraints. Matching values prove a cost increase of $2 = p_1 \times 1$.

Reset requirement 1 to 8; change requirement 2 to 3.6

Primal · second target: 3 → 3.6

$$\begin{aligned} \min \quad & 13x_1 + 10x_2 + 6x_3 \\ & 5x_1 + x_2 + 3x_3 = 8, \\ & 3x_1 + x_2 = 3.6, \qquad x \geq 0. \end{aligned}$$

Dual

$$\begin{aligned} \max \quad & 8p_1 + 3.6p_2 \\ & 5p_1 + 3p_2 \leq 13, \\ & p_1 + p_2 \leq 10, \quad 3p_1 \leq 6, \\ & p_1, p_2 \in \mathbb{R}. \end{aligned}$$

$$x_2 = 0: \quad 3x_1 = 3.6 \Rightarrow x_1 = 6/5, \quad 5(6/5) + 3x_3 = 8 \Rightarrow x_3 = 2/3.$$

$$13(6/5) + 6(2/3) = 19.6 = 8(2) + 3.6(1).$$

The minimum cost rises by 0.6, exactly $p_2^* \times 0.6$. This change is separate from the preceding one.

Marginal cost means the change in best cost per extra unit

Primal

$$\min \; c^\top x \quad \text{s.t. } Ax = b, \quad x \geq 0.$$

Dual

$$\max \; b^\top p \quad \text{s.t. } A^\top p \leq c, \quad p \in \mathbb{R}^m.$$

Keep A, c fixed. While the same optimal basis remains feasible, let x_{old} and x_{new} be its optimal solutions at b and $b + \Delta b$. Then

$$c^\top x_{\text{new}} - c^\top x_{\text{old}} = (p^*)^\top \Delta b.$$

As long as the same basis remains feasible, p_i^* gives the change in minimum cost per unit increase in the right-hand side of constraint i. This is its marginal cost. For equality constraints, it may be positive, zero, or negative.

After any RHS change, the old dual point still gives a lower bound. Equality with the new optimum needs a feasible solution attaining it.

Check quantities before trusting the price prediction

Primal · second target: 3 + ε

$$\begin{aligned} \min \quad & 13x_1 + 10x_2 + 6x_3 \\ & 5x_1 + x_2 + 3x_3 = 8, \\ & 3x_1 + x_2 = 3 + \varepsilon, \qquad x \geq 0. \end{aligned}$$

Dual

$$\begin{aligned} \max \quad & 8p_1 + (3+\varepsilon)p_2 \\ & 5p_1 + 3p_2 \leq 13, \\ & p_1 + p_2 \leq 10, \quad 3p_1 \leq 6, \\ & p_1, p_2 \in \mathbb{R}. \end{aligned}$$

Using the old basis, set $x_2 = 0$: $x_1 = 1 + \varepsilon/3$, $x_3 = 1 - 5\varepsilon/9$.

ε	x_1	x_3	**Old basis feasible?**
0	1	1	Yes
$3/5$	$6/5$	$2/3$	Yes
$9/5$	$8/5$	0	Yes: boundary
2	$5/3$	$-1/9$	No

Both quantities are nonnegative exactly when $-3 \leq \varepsilon \leq 9/5$. Optional slider and range derivation.

A larger change preserves the bound but loses primal feasibility

Primal · second target: 3 → 5

$$\begin{aligned} \min \quad & 13x_1 + 10x_2 + 6x_3 \\ & 5x_1 + x_2 + 3x_3 = 8, \\ & 3x_1 + x_2 = 5, \qquad x \geq 0. \end{aligned}$$

Dual

$$\begin{aligned} \max \quad & 8p_1 + 5p_2 \\ & 5p_1 + 3p_2 \leq 13, \\ & p_1 + p_2 \leq 10, \quad 3p_1 \leq 6, \\ & p_1, p_2 \in \mathbb{R}. \end{aligned}$$

$$x_1 = \frac{5}{3} - \frac{1}{3}x_2, \qquad x_3 = -\frac{1}{9} + \frac{2}{9}x_2, \qquad f = 21 + 7x_2.$$

At $x_2 = 0$, the basic value $x_3 = -1/9$ is infeasible. The reduced cost is still 7, and $p = (2, 1)$ still gives the lower bound 21.

Dual simplex repairs this mismatch. We will return to this dictionary after learning one pivot.

From a primal dictionary to a dual feasibility check

Primal

$$\min\ c^\top x \quad \text{s.t. } Ax = b, \quad x \geq 0.$$

Dual

$$\max\ b^\top p \quad \text{s.t. } A^\top p \leq c, \quad p \in \mathbb{R}^m.$$

Let A_j be column j of A, and c_j the original cost of x_j. To eliminate basic variables from the objective, choose p by solving

$$A_j^\top p = c_j \qquad \text{for every basic variable } x_j.$$

The independent basic columns determine this dual candidate. Basic reduced costs are zero.

Each nonbasic objective coefficient is the slack in its dual constraint:

$$\bar{c}_j = c_j - A_j^\top p; \qquad A_j^\top p \leq c_j \iff \bar{c}_j \geq 0.$$

All nonbasic reduced costs nonnegative ⇒ the associated dual point is feasible.

This construction does not require primal feasibility or complementary slackness.

Example 4.6: check the dual point from the basis

Primal · second target: 3 → 5

$$\begin{aligned} \min \quad & 13x_1 + 10x_2 + 6x_3 \\ & 5x_1 + x_2 + 3x_3 = 8, \\ & 3x_1 + x_2 = 5, \qquad x \geq 0. \end{aligned}$$

Dual

$$\begin{aligned} \max \quad & 8p_1 + 5p_2 \\ & 5p_1 + 3p_2 \leq 13, \\ & p_1 + p_2 \leq 10, \quad 3p_1 \leq 6, \\ & p_1, p_2 \in \mathbb{R}. \end{aligned}$$

Use the basic variables x_1, x_3, even though their current values are $5/3, -1/9$. Their columns give

$$5p_1 + 3p_2 = 13, \qquad 3p_1 = 6 \qquad \Longrightarrow \qquad p = (2, 1).$$

Variable	Dual left side at this point	Reduced cost = cost – left side
x_1 (basic)	$5(2) + 3(1) = 13$	$13 - 13 = 0$
x_2 (nonbasic)	$2 + 1 = 3$	$10 - 3 = 7$
x_3 (basic)	$3(2) = 6$	$6 - 6 = 0$

$f = 21 + 7x_2$ has nonnegative reduced costs: dual feasible. But $x_3 = -1/9$: primal infeasible.

Primal and dual simplex start from different conditions

Primal

$$\min \; c^\top x \quad \text{s.t. } Ax = b, \quad x \geq 0.$$

Dual

$$\max \; b^\top p \quad \text{s.t. } A^\top p \leq c, \quad p \in \mathbb{R}^m.$$

Condition	Test for the current basis
1. Primal feasibility	All basic values are nonnegative.
2. Dual feasibility	All reduced costs $\bar{c}_j = c_j - A_j^\top p$ are nonnegative.

Primal simplex starts with Condition 1 and works toward Condition 2.

Keep basic values nonnegative; seek nonnegative reduced costs.

Dual simplex starts with Condition 2 and works toward Condition 1.

Keep reduced costs nonnegative; seek nonnegative basic values.

When both conditions hold, the current basic solution is optimal.

Example 4.7: begin with its full dictionary

Primal · Example 4.7

$$\begin{aligned}\min \quad & 2x_1 + 6x_2 + 10x_3 \\ & -2x_1 + 4x_2 + x_3 + x_4 = 2, \\ & 4x_1 - 2x_2 - 3x_3 + x_5 = -1, \\ & x_1, x_2, x_3, x_4, x_5 \geq 0.\end{aligned}$$

Dual

$$\begin{aligned}\max \quad & 2p_1 - p_2 \\ & -2p_1 + 4p_2 \leq 2, \\ & 4p_1 - 2p_2 \leq 6, \\ & p_1 - 3p_2 \leq 10, \\ & p_1 \leq 0, \quad p_2 \leq 0.\end{aligned}$$

$$\begin{aligned}x_4 &= 2 + 2x_1 - 4x_2 - x_3, \\ x_5 &= -1 - 4x_1 + 2x_2 + 3x_3, \\ f &= 2x_1 + 6x_2 + 10x_3.\end{aligned}$$

Set $x_1 = x_2 = x_3 = 0$: $x_4 = 2$, $x_5 = -1$. Reduced costs $(2, 6, 10)$ are nonnegative, but x_5 is negative: choose x_5 to leave.

In the x_5 equation, increasing x_2 or x_3 can offset -1. Increasing x_1 makes it worse. Thus only x_2, x_3 may enter.

Try entering x2: the reduced costs stay nonnegative

Primal · Example 4.7

$$\begin{aligned}\min\quad & 2x_1+6x_2+10x_3\\ & -2x_1+4x_2+x_3+x_4=2,\\ & 4x_1-2x_2-3x_3+x_5=-1,\\ & x_1,x_2,x_3,x_4,x_5\ge 0.\end{aligned}$$

Dual

$$\begin{aligned}\max\quad & 2p_1-p_2\\ & -2p_1+4p_2\le 2,\\ & 4p_1-2p_2\le 6,\\ & p_1-3p_2\le 10,\\ & p_1\le 0,\quad p_2\le 0.\end{aligned}$$

Solve $x_5=-1-4x_1+2x_2+3x_3$ for x_2, then substitute into the objective:

$$\begin{aligned}x_2 &= \tfrac{1}{2}+2x_1-\tfrac{3}{2}x_3+\tfrac{1}{2}x_5,\\ f &= 2x_1+6(\tfrac{1}{2}+2x_1-\tfrac{3}{2}x_3+\tfrac{1}{2}x_5)+10x_3\\ &= 3+14x_1+x_3+3x_5.\end{aligned}$$

The new coefficient of x_3 is $10-\frac{6}{2}(3)=1$. All new reduced costs $(14,1,3)$ are nonnegative: dual feasibility is preserved.

Try entering x3: a reduced cost becomes negative

Primal · Example 4.7

$$\begin{aligned}\min\quad & 2x_1+6x_2+10x_3\\ & -2x_1+4x_2+x_3+x_4=2,\\ & 4x_1-2x_2-3x_3+x_5=-1,\\ & x_1,x_2,x_3,x_4,x_5\ge 0.\end{aligned}$$

Dual

$$\begin{aligned}\max\quad & 2p_1-p_2\\ & -2p_1+4p_2\le 2,\\ & 4p_1-2p_2\le 6,\\ & p_1-3p_2\le 10,\\ & p_1\le 0,\quad p_2\le 0.\end{aligned}$$

$$x_3=\tfrac{1}{3}+\tfrac{4}{3}x_1-\tfrac{2}{3}x_2+\tfrac{1}{3}x_5.$$

In $f=2x_1+6x_2+10x_3$, the new coefficient of x_2 would be

$$6+10(-\tfrac{2}{3})=6-\frac{10}{3}(2)=-\tfrac{2}{3}<0.$$

This choice loses dual feasibility.

The **dual ratio test** compares $6/2=3<10/3$, so x_2 enters.

Complete the chosen pivot: x2 enters and x5 leaves

Primal · Example 4.7

$$\begin{aligned}\min\quad & 2x_1+6x_2+10x_3\\ & -2x_1+4x_2+x_3+x_4=2,\\ & 4x_1-2x_2-3x_3+x_5=-1,\\ & x_1,x_2,x_3,x_4,x_5\ge 0.\end{aligned}$$

Dual

$$\begin{aligned}\max\quad & 2p_1-p_2\\ & -2p_1+4p_2\le 2,\\ & 4p_1-2p_2\le 6,\\ & p_1-3p_2\le 10,\\ & p_1\le 0,\quad p_2\le 0.\end{aligned}$$

Use the x_2 equation to substitute into the remaining x_4 row. The completed dictionary is

$$\begin{aligned}x_2 &= \tfrac{1}{2}+2x_1-\tfrac{3}{2}x_3+\tfrac{1}{2}x_5,\\ x_4 &= 2+2x_1-4(\tfrac{1}{2}+2x_1-\tfrac{3}{2}x_3+\tfrac{1}{2}x_5)-x_3\\ &= -6x_1+5x_3-2x_5,\\ f &= 3+14x_1+x_3+3x_5.\end{aligned}$$

Basic variables: x_2, x_4. Nonbasic variables: x_1, x_3, x_5.

The new dictionary passes both feasibility tests

Primal · Example 4.7

$$\begin{aligned}\min \quad & 2x_1 + 6x_2 + 10x_3 \\ & -2x_1 + 4x_2 + x_3 + x_4 = 2, \\ & 4x_1 - 2x_2 - 3x_3 + x_5 = -1, \\ & x_1, x_2, x_3, x_4, x_5 \geq 0.\end{aligned}$$

Dual

$$\begin{aligned}\max \quad & 2p_1 - p_2 \\ & -2p_1 + 4p_2 \leq 2, \\ & 4p_1 - 2p_2 \leq 6, \\ & p_1 - 3p_2 \leq 10, \\ & p_1 \leq 0, \quad p_2 \leq 0.\end{aligned}$$

Optimality condition	Check in the new dictionary
1. Primal feasibility	Set $x_1 = x_3 = x_5 = 0$: $x_2 = 1/2,\ x_4 = 0$. All values are nonnegative.
2. Dual feasibility	Reduced costs $(\bar{c}_1, \bar{c}_3, \bar{c}_5) = (14, 1, 3)$ are nonnegative.

Basic x_4, x_2 give $p_1 = 0,\ 4p_1 - 2p_2 = 6$, hence $p = (0, -3)$.

> Both solutions are feasible and their values match: $f = 6(1/2) = 3 = 2(0) - (-3)$. Thus both are optimal.

The zero basic value $x_4 = 0$ is allowed; the primal solution is degenerate.

The completed pivot in tableau notation

Primal · Example 4.7

$$\begin{aligned}\min\quad & 2x_1 + 6x_2 + 10x_3 \\ & -2x_1 + 4x_2 + x_3 + x_4 = 2, \\ & 4x_1 - 2x_2 - 3x_3 + x_5 = -1, \\ & x_1, x_2, x_3, x_4, x_5 \geq 0.\end{aligned}$$

Dual

$$\begin{aligned}\max\quad & 2p_1 - p_2 \\ & -2p_1 + 4p_2 \leq 2, \\ & 4p_1 - 2p_2 \leq 6, \\ & p_1 - 3p_2 \leq 10, \\ & p_1 \leq 0, \quad p_2 \leq 0.\end{aligned}$$

Row	Column 0	x_1	x_2	x_3	x_4	x_5
R_0	-3	14	0	1	0	3
x_4	0	6	0	-5	1	2
x_2	$1/2$	-2	1	$3/2$	0	$-1/2$

Move variables left in the constraint rows. Column 0 stores the RHS; in R_0 it stores $-f_0 = -3$, so the objective value is 3.

The row positions stay fixed: the former x_5 row now represents x_2. Initial tableau and the same row operations.

The dual ratio test keeps reduced costs nonnegative

Start with nonnegative reduced costs. Any negative basic value may leave; in our examples, choose the smallest variable index among them.

$$x_k = \beta + \sum_{j \in \mathcal{N}} \alpha_j x_j, \quad \beta < 0; \qquad f = f_0 + \sum_{j \in \mathcal{N}} \bar{c}_j x_j, \quad \bar{c}_j \geq 0.$$

$$x_k \text{ leaves}; \qquad j \in \arg\min_{i:\alpha_i > 0} \frac{\bar{c}_i}{\alpha_i} \text{ enters}.$$

Only a positive RHS coefficient can offset the negative constant. In an equality tableau, that coefficient is $v_i = -\alpha_i < 0$, so the ratio is $\bar{c}_i / |v_i|$.

Substitution keeps every new reduced cost nonnegative. Complete algebraic proof.

Repeat until one of the two stopping cases occurs

Step	Operation
Start	Check that every reduced cost is nonnegative.
Inspect basic values	If all are nonnegative, stop: both solutions are optimal.
Select the leaving row	Any negative basic value is eligible. Our convention: choose the smallest variable index among them.
Select the entering variable	Use the smallest reduced-cost / positive-RHS-coefficient ratio.
No eligible entering variable?	Stop: the primal is infeasible. Otherwise pivot and check the basic values again.

> The dual objective does not decrease; a zero ratio can leave it unchanged. The leaving convention alone does not prevent cycling. We study an anticycling rule later.

A negative row with no eligible entry proves infeasibility

$$x_k = -1 - 2x_2 - 3x_4, \qquad x_k, x_2, x_4 \geq 0.$$

The right side is at most –1 for every allowed choice. It cannot equal the nonnegative left side.

Generally, $\beta < 0$ and all $\alpha_j \leq 0$ make $x_k = \beta + \sum_j \alpha_j x_j < 0$ unavoidable.

The primal is infeasible. If the current dictionary is dual feasible, the dual is unbounded above.

Proof and the associated improving dual ray.

Return to the changed requirement: now select its pivot

Primal · second target: 3 → 5

$$\begin{aligned}\min \quad & 13x_1 + 10x_2 + 6x_3 \\ & 5x_1 + x_2 + 3x_3 = 8, \\ & 3x_1 + x_2 = 5, \qquad x \geq 0.\end{aligned}$$

Dual

$$\begin{aligned}\max \quad & 8p_1 + 5p_2 \\ & 5p_1 + 3p_2 \leq 13, \\ & p_1 + p_2 \leq 10, \quad 3p_1 \leq 6, \\ & p_1, p_2 \in \mathbb{R}.\end{aligned}$$

$$x_1 = \frac{5}{3} - \frac{1}{3}x_2, \qquad x_3 = -\frac{1}{9} + \frac{2}{9}x_2, \qquad f = 21 + 7x_2.$$

x_3 leaves. The only eligible variable is x_2, with ratio $7/(2/9) = 63/2$. Solve its row: $x_2 = 1/2 + (9/2)x_3$.

We can reuse the old basis and repair it without restarting Phase I.

The repaired basis attains the new minimum cost

Primal · second target: 3 → 5

$$\begin{aligned}\min \quad & 13x_1 + 10x_2 + 6x_3 \\ & 5x_1 + x_2 + 3x_3 = 8, \\ & 3x_1 + x_2 = 5, \qquad x \geq 0.\end{aligned}$$

Dual

$$\begin{aligned}\max \quad & 8p_1 + 5p_2 \\ & 5p_1 + 3p_2 \leq 13, \\ & p_1 + p_2 \leq 10, \quad 3p_1 \leq 6, \\ & p_1, p_2 \in \mathbb{R}.\end{aligned}$$

$$x_1 = \tfrac{3}{2} - \tfrac{3}{2}x_3, \qquad x_2 = \tfrac{1}{2} + \tfrac{9}{2}x_3, \qquad f = \tfrac{49}{2} + \tfrac{63}{2}x_3.$$

Basic x_1, x_2 make $5p_1 + 3p_2 = 13$, $p_1 + p_2 = 10$, giving $p = (-17/2, 37/2)$.

Both feasible points attain $49/2 = 24.5$: $x = (3/2, 1/2, 0)$, and $8p_1 + 5p_2 = 49/2$. The old lower bound 21 was valid but not attained.

Recap: preserve dual feasibility, repair primal feasibility

For a minimization dictionary, dual simplex keeps every reduced cost nonnegative and works toward nonnegative basic values.

$$x_k = \beta + \sum_{j \in \mathcal{N}} \alpha_j x_j, \quad \beta < 0; \qquad f = f_0 + \sum_{j \in \mathcal{N}} \bar{c}_j x_j, \quad \bar{c}_j \geq 0.$$

Here $\mathcal{N}$ lists the nonbasic variables, and α_j is the coefficient on the **right-hand side** of the chosen row.

Operation	What to do
Leave	Choose a basic variable with a negative value. Our convention: smallest variable index.
Enter	Among $\alpha_j > 0$, minimize $\bar{c}_j / \alpha_j$. This preserves nonnegative reduced costs.
Pivot	Solve the chosen row for the entering variable; substitute into every other row and the objective.

Stop: all basic values nonnegative ⇒ optimal. A negative row with no positive RHS coefficient ⇒ primal infeasible.

Example 4.8: follow the same basis in two spaces

We know how to perform a dual-simplex pivot. What does it do in the two plots?

In this example, the dual points remain feasible, while the primal points become feasible at the final pivot.

Primal · Example 4.8

$$\begin{aligned} \min \quad & x_1 + x_2 \\ & x_1 + 2x_2 \geq 2, \quad x_1 \geq 1, \\ & x_1, x_2 \geq 0. \end{aligned}$$

Dual

$$\begin{aligned} \max \quad & 2p_1 + p_2 \\ & p_1 + p_2 \leq 1, \quad 2p_1 \leq 1, \\ & p_1, p_2 \geq 0. \end{aligned}$$

For the pivots, use the standard-form equalities $x_1 + 2x_2 - x_3 = 2$, $x_1 - x_4 = 1$, with $x_3, x_4 \geq 0$.

Each basis sets nonbasic primal variables to zero and makes the corresponding dual inequalities active.

Basic surplus columns give $-p_1 = 0$ or $-p_2 = 0$.

We will calculate each dictionary and plot its primal and dual points. Matching letters identify the same basis.

Basis A: dual feasible, with two negative basic values

Primal · Example 4.8

$$\begin{aligned} \min \quad & x_1 + x_2 \\ & x_1 + 2x_2 \geq 2, \quad x_1 \geq 1, \\ & x_1, x_2 \geq 0. \end{aligned}$$

Dual

$$\begin{aligned} \max \quad & 2p_1 + p_2 \\ & p_1 + p_2 \leq 1, \quad 2p_1 \leq 1, \\ & p_1, p_2 \geq 0. \end{aligned}$$

$$\begin{aligned} x_3 &= -2 + x_1 + 2x_2, \\ x_4 &= -1 + x_1, \\ f &= x_1 + x_2. \end{aligned}$$

$$\mathcal{B} = (x_3, x_4): \quad -p_1 = 0, \quad -p_2 = 0 \quad \Rightarrow \quad p = (0, 0).$$

Both $x_3 = -2$ and $x_4 = -1$ are negative. Either may leave; our smallest-index convention selects x_3.

Using the x_3 row, ratios are $1/1 = 1$ for x_1 and $1/2$ for x_2. Thus x_2 enters.

Locate basis A before the first pivot

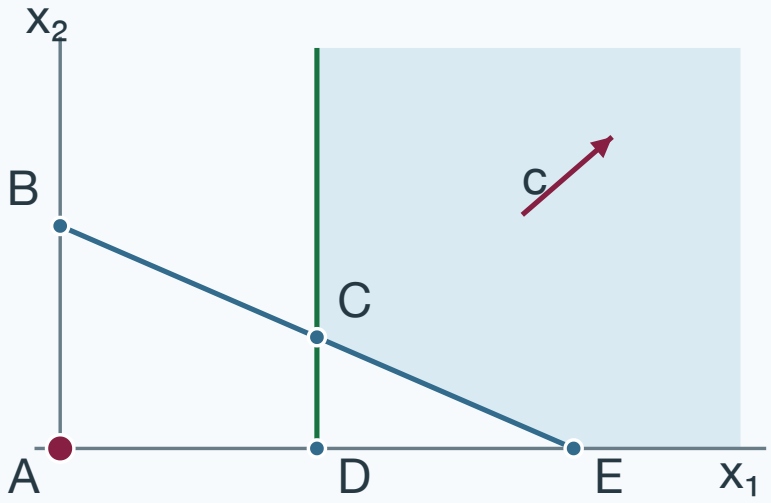
Primal · Example 4.8

$$\begin{aligned} \min \quad & x_1 + x_2 \\ & x_1 + 2x_2 \geq 2, \quad x_1 \geq 1, \\ & x_1, x_2 \geq 0. \end{aligned}$$

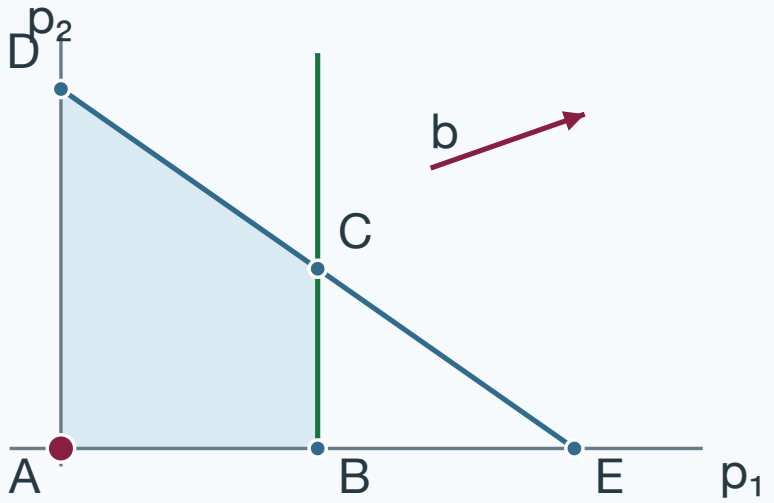
Dual

$$\begin{aligned} \max \quad & 2p_1 + p_2 \\ & p_1 + p_2 \leq 1, \quad 2p_1 \leq 1, \\ & p_1, p_2 \geq 0. \end{aligned}$$

Primal variables



Dual variables



Blue: feasible region (clipped). Orange: pivot path. Arrows: increasing objective.

A : $x = (0, 0, -2, -1)$, $p = (0, 0)$, objective 0. Primal inequalities fail; the dual point lies in its feasible region.

Pivot 1 gives basis B, but one negative value remains

Primal · Example 4.8

$$\begin{aligned} \min \quad & x_1 + x_2 \\ & x_1 + 2x_2 \geq 2, \quad x_1 \geq 1, \\ & x_1, x_2 \geq 0. \end{aligned}$$

Dual

$$\begin{aligned} \max \quad & 2p_1 + p_2 \\ & p_1 + p_2 \leq 1, \quad 2p_1 \leq 1, \\ & p_1, p_2 \geq 0. \end{aligned}$$

$$\begin{aligned} x_2 &= 1 - \tfrac{1}{2}x_1 + \tfrac{1}{2}x_3, \\ x_4 &= -1 + x_1, \\ f &= 1 + \tfrac{1}{2}x_1 + \tfrac{1}{2}x_3. \end{aligned}$$

$$\mathcal{B} = (x_2, x_4): \quad 2p_1 = 1, \quad -p_2 = 0 \quad \Rightarrow \quad p = (1/2, 0).$$

The point is $x = (0, 1, 0, -1)$, value 1. Next $x_4 = -1 + x_1$ must leave; only x_1 can enter.

A to B improves the bound; primal feasibility still fails

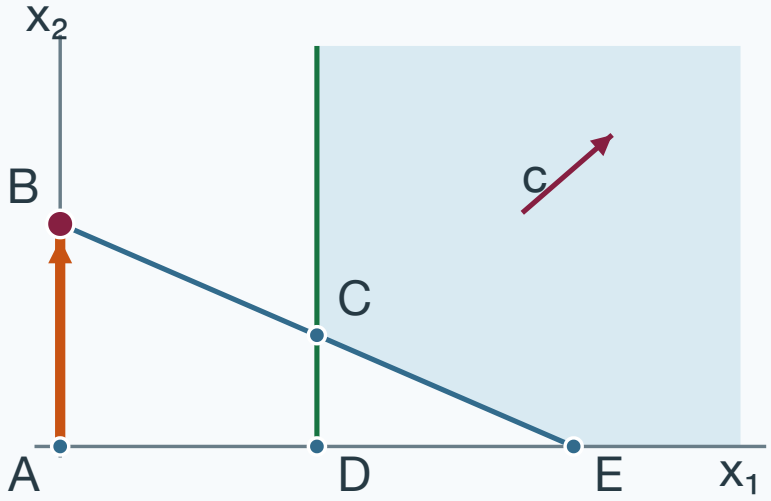
Primal · Example 4.8

$$\begin{aligned} \min \quad & x_1 + x_2 \\ & x_1 + 2x_2 \geq 2, \quad x_1 \geq 1, \\ & x_1, x_2 \geq 0. \end{aligned}$$

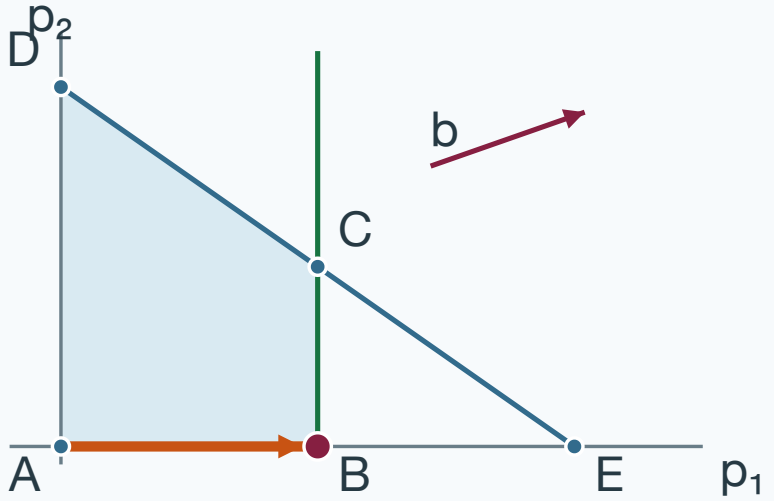
Dual

$$\begin{aligned} \max \quad & 2p_1 + p_2 \\ & p_1 + p_2 \leq 1, \quad 2p_1 \leq 1, \\ & p_1, p_2 \geq 0. \end{aligned}$$

Primal variables



Dual variables



Blue: feasible region (clipped). Orange: pivot path. Arrows: increasing objective.

B : $(x_1, x_2) = (0, 1), p = (1/2, 0)$. The dual bound increases to 1; the primal row $x_1 \geq 1$ still fails.

Pivot 2 gives basis C and passes both tests

Primal · Example 4.8

$$\begin{aligned} \min \quad & x_1 + x_2 \\ & x_1 + 2x_2 \geq 2, \quad x_1 \geq 1, \\ & x_1, x_2 \geq 0. \end{aligned}$$

Dual

$$\begin{aligned} \max \quad & 2p_1 + p_2 \\ & p_1 + p_2 \leq 1, \quad 2p_1 \leq 1, \\ & p_1, p_2 \geq 0. \end{aligned}$$

$$\begin{aligned} x_1 &= 1 + x_4, \\ x_2 &= \tfrac{1}{2} + \tfrac{1}{2}x_3 - \tfrac{1}{2}x_4, \\ f &= \tfrac{3}{2} + \tfrac{1}{2}x_3 + \tfrac{1}{2}x_4. \end{aligned}$$

$$\mathcal{B} = (x_1, x_2): \quad p_1 + p_2 = 1, \quad 2p_1 = 1 \quad \Rightarrow \quad p = (1/2, 1/2).$$

At $x_3 = x_4 = 0$, $x = (1, 1/2, 0, 0)$ and $p = (1/2, 1/2)$ are feasible with equal value $3/2$.

C is feasible in both pictures and attains the matching bound

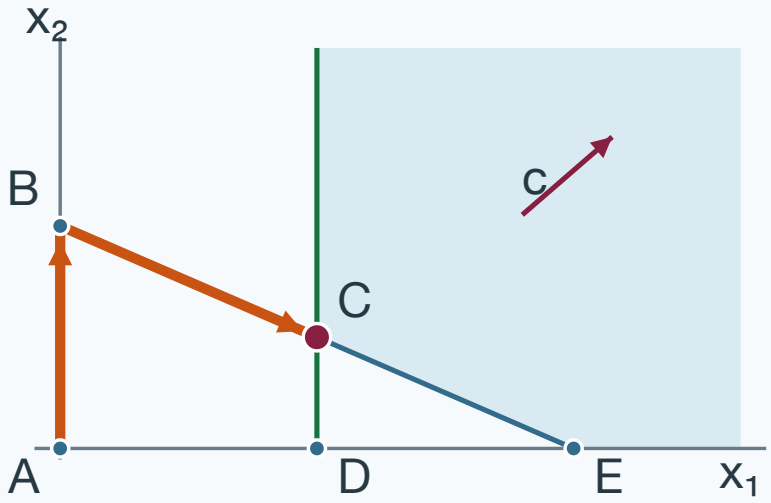
Primal · Example 4.8

$$\begin{aligned} \min \quad & x_1 + x_2 \\ & x_1 + 2x_2 \geq 2, \quad x_1 \geq 1, \\ & x_1, x_2 \geq 0. \end{aligned}$$

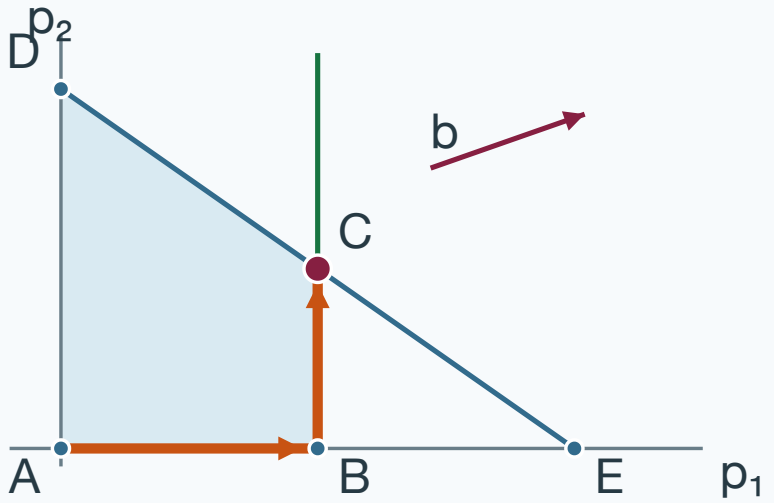
Dual

$$\begin{aligned} \max \quad & 2p_1 + p_2 \\ & p_1 + p_2 \leq 1, \quad 2p_1 \leq 1, \\ & p_1, p_2 \geq 0. \end{aligned}$$

Primal variables



Dual variables



Blue: feasible region (clipped). Orange: pivot path. Arrows: increasing objective.

The dual-simplex route is A → B → C. Labels D and E show other bases, not additional pivots in this run.

Complete five-basis correspondence.

The dual points are adjacent or coincide

> The dual-simplex algorithm stays dual feasible, with a nondecreasing dual objective. A pivot changes one basic column.

Consecutive dual bases share all but one of their independent active equalities. Their points are adjacent, or they coincide in a degenerate step.

> The same basis changes are represented by dictionaries of the primal equations. Their basic quantities can be negative until the final step.

There is no need to convert the dual into standard form and build a separate simplex tableau.

Practice: choose a dual-simplex pivot

Primal

$$\begin{aligned}\min \quad & 3x_1 + 2x_2 \\ & -x_1 + 2x_2 + x_3 = 4, \\ & -x_1 - 4x_2 + x_4 = -2, \qquad x \geq 0.\end{aligned}$$

Dual

$$\begin{aligned}\max \quad & 4p_1 - 2p_2 \\ & -p_1 - p_2 \leq 3, \qquad 2p_1 - 4p_2 \leq 2, \\ & p_1, p_2 \leq 0.\end{aligned}$$

$$x_3 = 4 + x_1 - 2x_2, \qquad x_4 = -2 + x_1 + 4x_2, \qquad f = 3x_1 + 2x_2.$$

Identify the leaving variable. Compare the eligible ratios, perform the pivot, and check whether another pivot is needed.

The next page gives the answer. Check the basic values and the reduced costs separately.

Practice answer: the new basis is optimal

Primal

$$\begin{aligned}\min \quad & 3x_1 + 2x_2 \\ & -x_1 + 2x_2 + x_3 = 4, \\ & -x_1 - 4x_2 + x_4 = -2, \quad x \geq 0.\end{aligned}$$

Dual

$$\begin{aligned}\max \quad & 4p_1 - 2p_2 \\ & -p_1 - p_2 \leq 3, \quad 2p_1 - 4p_2 \leq 2, \\ & p_1, p_2 \leq 0.\end{aligned}$$

x_4 leaves. Ratios $3/1 = 3$, $2/4 = 1/2$ select x_2.

$$x_2 = \tfrac{1}{2} - \tfrac{1}{4}x_1 + \tfrac{1}{4}x_4, \quad x_3 = 3 + \tfrac{3}{2}x_1 - \tfrac{1}{2}x_4, \quad f = 1 + \tfrac{5}{2}x_1 + \tfrac{1}{2}x_4.$$

$x = (0, 1/2, 3, 0)$, $p = (0, -1/2)$: both feasible, both value 1. Nonnegative basic quantities and reduced costs justify stopping.

A zero nonbasic reduced cost means dual degeneracy

Primal

$$\min\ c^\top x \quad \text{s.t. } Ax = b, \quad x \geq 0.$$

Dual

$$\max\ b^\top p \quad \text{s.t. } A^\top p \leq c, \quad p \in \mathbb{R}^m.$$

The m basic columns determine p through $A_{B(i)}^\top p = c_{B(i)}$. These are m independent active dual constraints.

$$\text{For a nonbasic column } j: \quad \bar{c}_j = 0 \iff A_j^\top p = c_j.$$

> This additional active constraint makes the associated dual basic solution degenerate. A zero selected ratio can change the basis without improving the dual objective.

Do not confuse this with a zero basic primal variable. We will revisit that separate issue in Example 4.9.

Use a lexicographic column comparison to avoid cycling

Keep the objective row first and the constraint-row positions fixed. Choose a negative right-hand-side row.

> For each eligible column j, divide its **entire column** by $|v_j|$, where its coefficient in the chosen row is $v_j < 0$.

Compare normalized columns from top to bottom; choose the smaller first unequal entry. If the entire columns tie, choose the smallest variable index.

> The first entry is the ordinary dual ratio. Later entries resolve a ratio tie.

The finite-termination guarantee also needs initially lexicographically positive variable columns. We check that condition after a genuine tie example.

A modified cost creates an actual ratio tie

Primal · variant: cost of x_3 is 9

$$\begin{aligned}\min\quad & 2x_1 + 6x_2 + 9x_3 \\ & -2x_1 + 4x_2 + x_3 + x_4 = 2, \\ & 4x_1 - 2x_2 - 3x_3 + x_5 = -1, \\ & x_1, x_2, x_3, x_4, x_5 \geq 0.\end{aligned}$$

Dual

$$\begin{aligned}\max\quad & 2p_1 - p_2 \\ & -2p_1 + 4p_2 \leq 2, \\ & 4p_1 - 2p_2 \leq 6, \\ & p_1 - 3p_2 \leq 9, \\ & p_1 \leq 0, \quad p_2 \leq 0.\end{aligned}$$

Row	Column 0	x_1	x_2	x_3	x_4	x_5
R_0	0	2	6	9	0	0
x_4	2	−2	4	1	1	0
x_5	−1	4	−2	−3	0	1

Choose the negative x_5 row. The ordinary ratios tie: $6/2 = 9/3 = 3$. Fixed row order is (R_0, x_4, x_5).

The next entry breaks the tie: variable 3 enters

Primal · variant: cost of x_3 is 9

$$\begin{aligned}\min \quad & 2x_1 + 6x_2 + 9x_3 \\ & -2x_1 + 4x_2 + x_3 + x_4 = 2, \\ & 4x_1 - 2x_2 - 3x_3 + x_5 = -1, \\ & x_1, x_2, x_3, x_4, x_5 \geq 0.\end{aligned}$$

Dual

$$\begin{aligned}\max \quad & 2p_1 - p_2 \\ & -2p_1 + 4p_2 \leq 2, \\ & 4p_1 - 2p_2 \leq 6, \\ & p_1 - 3p_2 \leq 9, \\ & p_1 \leq 0, \quad p_2 \leq 0.\end{aligned}$$

$$\frac{C_2}{2} = \begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix}, \qquad \frac{C_3}{3} = \begin{pmatrix} 3 \\ 1/3 \\ -1 \end{pmatrix}.$$

Ratios tie at 3. The next entries give $1/3 < 2$: x_3 enters, x_5 leaves.

Complete pivot and the final dictionary.

Lexicographic positivity is the termination hypothesis

A column is lexicographically positive when its first nonzero entry is positive. In the tied example, the first three columns start with positive costs; the two slack columns first have a positive unit entry.

> **Finite-termination statement:** if every initial variable column is lexicographically positive, the dual lexicographic rule preserves this property and terminates after finitely many pivots.

This condition is not another entering-variable test. Eligibility still comes from a negative coefficient in the selected tableau row.

Complete proof: column positivity and strict lexicographic progress.

Example 4.9: a different issue, several bases at one primal point

Primal · Example 4.9

$$\begin{aligned} \min \quad & 3x_1 + x_2 \\ & x_1 + x_2 \geq 2, \quad 2x_1 - x_2 \geq 0, \\ & x_1, x_2 \geq 0. \end{aligned}$$

Dual

$$\begin{aligned} \max \quad & 2p_1 \\ & p_1 + 2p_2 \leq 3, \quad p_1 - p_2 \leq 1, \\ & p_1, p_2 \geq 0. \end{aligned}$$

Add surplus variables: $x_1 + x_2 - x_3 = 2$, $2x_1 - x_2 - x_4 = 0$.

$$(x_1, x_2, x_3, x_4) = (0, 0, -2, 0).$$

This primal basic solution is infeasible, and it has a basic zero. Three different bases represent this same point; their associated dual points need not coincide.

Compute the dual point from the chosen basic columns

Primal · Example 4.9

$$\begin{aligned} \min \quad & 3x_1 + x_2 \\ & x_1 + x_2 \geq 2, \quad 2x_1 - x_2 \geq 0, \\ & x_1, x_2 \geq 0. \end{aligned}$$

Dual

$$\begin{aligned} \max \quad & 2p_1 \\ & p_1 + 2p_2 \leq 3, \quad p_1 - p_2 \leq 1, \\ & p_1, p_2 \geq 0. \end{aligned}$$

Basic variables	Active dual equalities	Associated dual point
(x_3, x_4)	$-p_1 = 0, \ -p_2 = 0$	A: $(0, 0)$, feasible
(x_1, x_3)	$p_1 + 2p_2 = 3, \ -p_1 = 0$	A′: $(0, 3/2)$, feasible
(x_2, x_3)	$p_1 - p_2 = 1, \ -p_1 = 0$	A″: $(0, -1)$, infeasible

The primal point alone does not determine its associated dual solution. The basis matters.

One primal point A corresponds to three different dual points

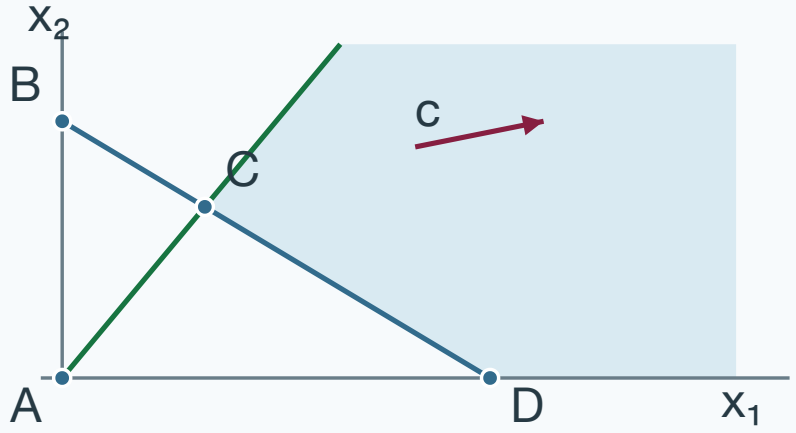
Primal · Example 4.9

$$\begin{aligned} \min \quad & 3x_1 + x_2 \\ & x_1 + x_2 \geq 2, \quad 2x_1 - x_2 \geq 0, \\ & x_1, x_2 \geq 0. \end{aligned}$$

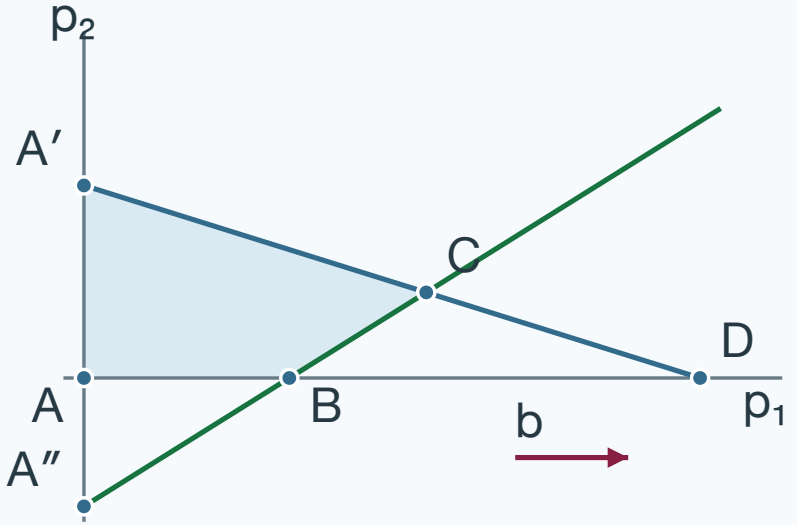
Dual

$$\begin{aligned} \max \quad & 2p_1 \\ & p_1 + 2p_2 \leq 3, \quad p_1 - p_2 \leq 1, \\ & p_1, p_2 \geq 0. \end{aligned}$$

Primal variables



Dual variables



Blue: feasible region (clipped). Arrows: increasing objective.

A, A′, A″ come from the three bases just calculated. A″ fails the dual sign restriction; the other two are feasible but not optimal.

Dual simplex either repairs feasibility or proves it impossible

Successful stop: basic values and reduced costs are nonnegative. The associated primal and dual solutions attain equal objective values.

Failed repair: a negative basic row has no eligible entering column. Its equation certifies primal infeasibility.

Farkas' lemma develops the general theory of such infeasibility certificates in the next lecture.

Study guide: prices, pivots, and two kinds of degeneracy

Be able to do	Core connection
Interpret a dual variable and check the range	Marginal cost and a feasible basis
Choose and perform a dual-simplex pivot	Preserve nonnegative reduced costs
Explain both stopping cases	Matching bounds or an impossible row
Follow primal and dual points in the same run	Example 4.8
Distinguish the degeneracy phenomena	Zero reduced cost versus zero basic primal value

The appendix retains the complete proofs, detailed range exploration, supporting-plane visualization, tableaux, and full basis tables.

Appendix: complete proofs and further exploration

Use these pages for the full algebra, optional geometry, and additional state comparisons.

- Marginal-cost slider and exact range
- Optional mixture and supporting planes
- Dual-simplex correctness and stopping cases
- Lexicographic finite termination
- Complete basis correspondence tables

Reduced cost is the extra cost after adjusting the basic ingredients

Primal · Example 4.6

$$\begin{aligned} \min \quad & 13x_1 + 10x_2 + 6x_3 \\ & 5x_1 + x_2 + 3x_3 = 8, \\ & 3x_1 + x_2 = 3, \qquad x \geq 0. \end{aligned}$$

Dual

$$\begin{aligned} \max \quad & 8p_1 + 3p_2 \\ & 5p_1 + 3p_2 \leq 13, \\ & p_1 + p_2 \leq 10, \quad 3p_1 \leq 6, \\ & p_1, p_2 \in \mathbb{R}. \end{aligned}$$

Treat a column as an ingredient's two contents. At $p = (2, 1)$, its weighted contents cost is $p^\top A_j$.

Ingredient	Contents A_j	Actual cost c_j	$p^\top A_j$	Gap $\bar{c}_j$
1	$(5, 3)$	13	13	0
2	$(1, 1)$	10	3	7
3	$(3, 0)$	6	6	0

The dictionary objective is $f = 19 + 7x_2$. The optimum uses ingredients 1 and 3; ingredient 2 has a positive extra cost. Optional geometric mixture example.

Optional exploration: when does a basic quantity become negative?

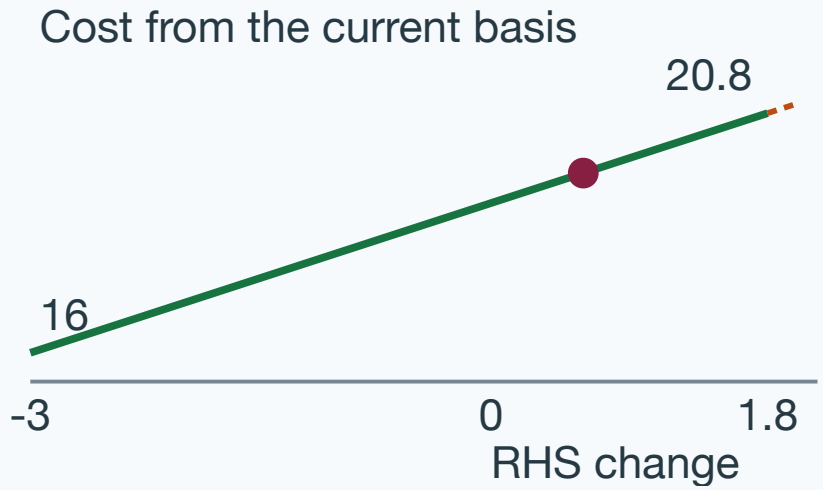
Primal · second target: 3 + ε

$$\begin{aligned} \min \quad & 13x_1 + 10x_2 + 6x_3 \\ & 5x_1 + x_2 + 3x_3 = 8, \\ & 3x_1 + x_2 = 3 + \varepsilon, \qquad x \geq 0. \end{aligned}$$

Dual

$$\begin{aligned} \max \quad & 8p_1 + (3+\varepsilon)p_2 \\ & 5p_1 + 3p_2 \leq 13, \\ & p_1 + p_2 \leq 10, \quad 3p_1 \leq 6, \\ & p_1, p_2 \in \mathbb{R}. \end{aligned}$$

Keep $x_2 = 0$: $x_1 = 1 + \varepsilon/3$, $x_3 = 1 - 5\varepsilon/9$. Cost $19 + \varepsilon$ is attained while $-3 \leq \varepsilon \leq 9/5$.



Change 0.6: quantities (1.2, 0, 0.667); cost 19.6. The current basis remains feasible and optimal.

Optional mixture example: the complete model

$$
\begin{aligned}
\min \quad & 7x_B + 4x_C + 4x_D + x_E + 4x_F \\
\text{s.t.} \quad & \tfrac{1}{2}x_B + 3x_C + 2x_E = 1, \quad \tfrac{1}{2}x_B + 3x_D + \tfrac{1}{2}x_E = 1, \\
& x_B + x_C + x_D + x_E + x_F = 1, \qquad x_B, x_C, x_D, x_E, x_F \geq 0.
\end{aligned}
$$

Column	B	C	D	E	F
First content	$1/2$	3	0	2	0
Second content	$1/2$	0	3	$1/2$	0
Cost	7	4	4	1	4

The two contents must total 1 each; the nonnegative mixture weights must sum to 1. A dual vector has three coordinates, one for each equality.

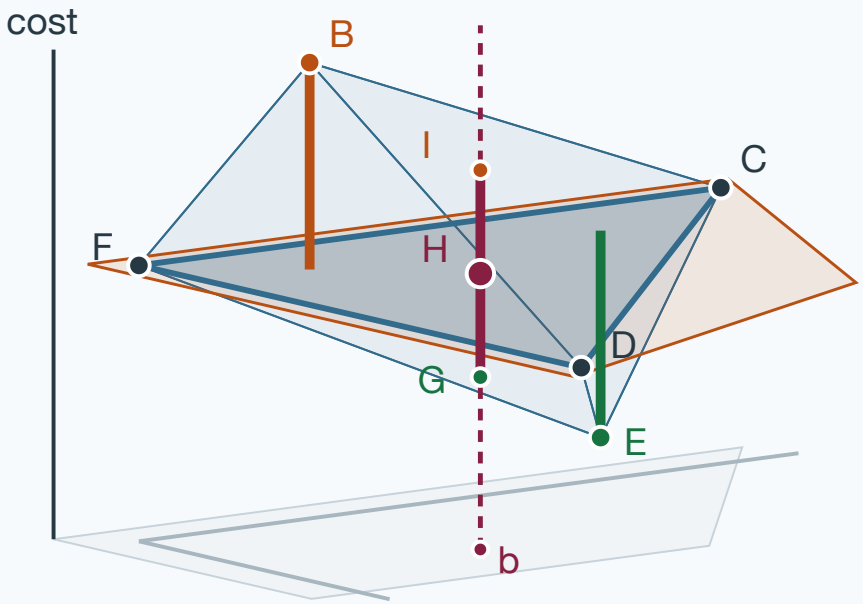
A plane matching the basic columns can fail another column

$$\begin{aligned} \min \quad & 7x_B + 4x_C + 4x_D + x_E + 4x_F \\ \text{s.t.} \quad & \tfrac{1}{2}x_B + 3x_C + 2x_E = 1, \quad \tfrac{1}{2}x_B + 3x_D + \tfrac{1}{2}x_E = 1, \\ & x_B + x_C + x_D + x_E + x_F = 1, \qquad x_B, x_C, x_D, x_E, x_F \geq 0. \end{aligned}$$

The mixture $x_C = x_D = x_F = 1/3$ has cost 4.

Their plane $z = 4$ corresponds to $p = (0, 0, 4)$: the last coordinate prices the sum-to-one row.

Column E costs $1 < 4$, so its dual inequality fails. Matching three columns does not prove a bound.



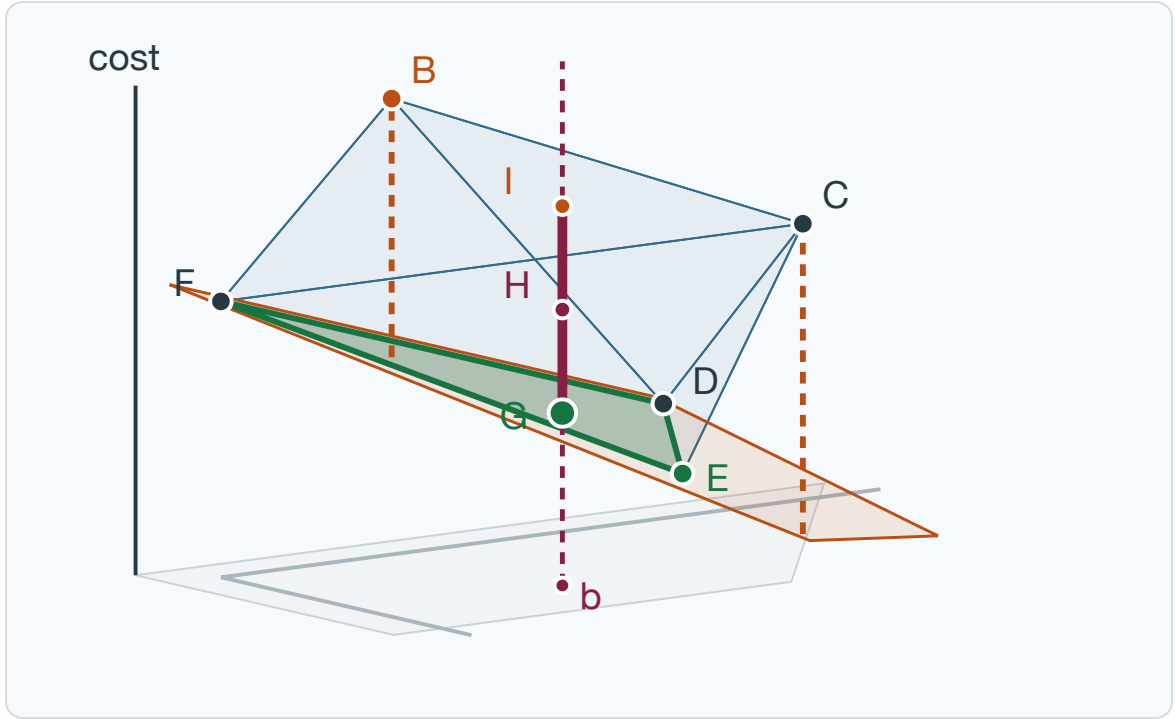
A plane below every column gives a valid lower bound

$$\begin{aligned} \min \quad & 7x_B + 4x_C + 4x_D + x_E + 4x_F \\ \text{s.t.} \quad & \tfrac{1}{2}x_B + 3x_C + 2x_E = 1, \quad \tfrac{1}{2}x_B + 3x_D + \tfrac{1}{2}x_E = 1, \\ & x_B + x_C + x_D + x_E + x_F = 1, \qquad x_B, x_C, x_D, x_E, x_F \geq 0. \end{aligned}$$

Use $p = (-3/2, 0, 4)$. At column contents (a_1, a_2), its weighted value is $z = 4 - 3a_1/2$.

> Every column lies on or above the plane. At required contents $(1, 1)$, the bound is $-3/2 + 0 + 4 = 5/2$.

The mixture $(x_D, x_E, x_F) = (1/4, 1/2, 1/4)$, with B and C unused, attains it.



Why these weighted column inequalities bound every mixture

$$\begin{aligned} \min \quad & 7x_B + 4x_C + 4x_D + x_E + 4x_F \\ \text{s.t.} \quad & \tfrac{1}{2}x_B + 3x_C + 2x_E = 1, \quad \tfrac{1}{2}x_B + 3x_D + \tfrac{1}{2}x_E = 1, \\ & x_B + x_C + x_D + x_E + x_F = 1, \qquad x_B, x_C, x_D, x_E, x_F \geq 0. \end{aligned}$$

For $p = (-3/2, 0, 4)$, every column $A_j = (a_{1j}, a_{2j}, 1)$ satisfies $c_j \geq p^\top A_j$. Multiply by $x_j \geq 0$ and add:

$$c^\top x \geq -\frac{3}{2}\sum_j a_{1j}x_j + 0\sum_j a_{2j}x_j + 4\sum_j x_j = -\frac{3}{2} + 4 = \frac{5}{2}.$$

D, E, F attain equality. B and C have reduced costs $15/4$ and $9/2$.

Calculate the range from nonnegative quantities

Primal · Example 4.6

$$\begin{aligned} \min \quad & 13x_1 + 10x_2 + 6x_3 \\ & 5x_1 + x_2 + 3x_3 = 8, \\ & 3x_1 + x_2 = 3 + \varepsilon, \qquad x \geq 0. \end{aligned}$$

Dual

$$\begin{aligned} \max \quad & 8p_1 + (3+\varepsilon)p_2 \\ & 5p_1 + 3p_2 \leq 13, \\ & p_1 + p_2 \leq 10, \quad 3p_1 \leq 6, \\ & p_1, p_2 \in \mathbb{R}. \end{aligned}$$

$$\begin{aligned} x_1 = 1 + \varepsilon/3 \geq 0 &\iff \varepsilon \geq -3, \\ x_3 = 1 - 5\varepsilon/9 \geq 0 &\iff \varepsilon \leq 9/5. \end{aligned}$$

With $x_2 = 0$, this basis is feasible for $-3 \leq \varepsilon \leq 9/5$. Reduced costs stay $(0, 7, 0)$, so the basis remains optimal.

At either endpoint a basic quantity is zero; in the interior both are positive.

Beyond the range, a new basis gives a new price

Primal · Example 4.6

$$\begin{aligned} \min \quad & 13x_1 + 10x_2 + 6x_3 \\ & 5x_1 + x_2 + 3x_3 = 8, \\ & 3x_1 + x_2 = 5, \qquad x \geq 0. \end{aligned}$$

Dual

$$\begin{aligned} \max \quad & 8p_1 + 5p_2 \\ & 5p_1 + 3p_2 \leq 13, \\ & p_1 + p_2 \leq 10, \quad 3p_1 \leq 6, \\ & p_1, p_2 \in \mathbb{R}. \end{aligned}$$

$$x_3 = 0 \quad \Longrightarrow \quad x = (3/2, 1/2, 0), \qquad c^\top x = 49/2.$$

$$5p_1 + 3p_2 = 13, \quad p_1 + p_2 = 10 \quad \Longrightarrow \quad p = (-17/2, 37/2).$$

The remaining dual inequality holds: $3p_1 = -51/2 \leq 6$. The dual value is $8p_1 + 5p_2 = 49/2$, matching the feasible primal cost.

The arithmetic has a compact matrix form

Let $v(b)$ be the minimum cost at requirements b, with A, c fixed.

For an optimal basis B with $B^{-1}b \geq 0$ and nonnegative reduced costs, the quantities and dual variables are

$$x_{\mathcal{B}} = B^{-1}b, \qquad (p^*)^\top = c_{\mathcal{B}}^\top B^{-1}.$$

$$\begin{aligned} c_{\mathcal{B}}^\top B^{-1}(b + \Delta b) &= c_{\mathcal{B}}^\top B^{-1}b + c_{\mathcal{B}}^\top B^{-1}\Delta b \\ &= v(b) + (p^*)^\top \Delta b. \end{aligned}$$

This is the optimum whenever $B^{-1}(b + \Delta b) \geq 0$, because the reduced costs do not depend on b.

Why nondegeneracy gives room for small changes

Assume $B^{-1}b > 0$. Every current basic quantity is strictly positive.

The changed quantities are $B^{-1}b + B^{-1}\Delta b$. For sufficiently small Δb, none can reach zero.

Thus the same basis stays feasible in a neighborhood, and the value is a linear function there. Its coefficient on requirement i is the optimal dual variable for that row.

If a basic quantity already equals zero, some arbitrarily small changes may make it negative. This explains why the nondegeneracy assumption matters for a two-sided neighborhood.

Why the minimum cost is piecewise linear and convex

Let $v(b)$ denote the minimum cost at requirements b.

With A, c fixed, let $D = \{p : A^\top p \le c\}$. At every feasible RHS with finite optimum, strong duality gives

$$v(b) = \max_{p \in D} b^\top p.$$

Each feasible dual solution supplies a linear lower bound. Their maximum is a convex, piecewise-linear value function on the region with finite value.

> Inside a region where one optimal basis remains feasible, its formula is linear. At a boundary, another dual solution can determine a different slope.

A finite value does not imply differentiability. An optimal dual vector remains a valid supporting lower bound even at a corner.

Proof: the unchanged basis gives the new optimum

Assume the basis B has $\bar{c} \geq 0$, and let $\hat{b} = b + \Delta b$ satisfy $B^{-1}\hat{b} \geq 0$.

$$\hat{x}_{\mathcal{B}} = B^{-1}\hat{b}, \quad \hat{x}_{\mathcal{N}} = 0, \qquad p^{\top} = c_{\mathcal{B}}^{\top} B^{-1}.$$

The first vector is primal feasible. Because A, c, B have not changed, $A^{\top} p \leq c$ remains true.

$$c^{\top}\hat{x} = c_{\mathcal{B}}^{\top} B^{-1}\hat{b} = p^{\top}\hat{b}.$$

Weak duality and equal objective values prove that both vectors are optimal. Subtract the corresponding old equality to obtain the marginal-cost formula.

Proof: substitute the pivot equation into the objective

Write the negative row as $x_k = \beta + \sum_i \alpha_i x_i$, with $\beta < 0$. Choose $\alpha_j > 0$. Solving gives

$$x_j = -\frac{\beta}{\alpha_j} + \frac{x_k}{\alpha_j} - \sum_{i \neq j} \frac{\alpha_i}{\alpha_j} x_i.$$

Substitute into $f = f_0 + \sum_i \bar{c}_i x_i$, and let $t = \bar{c}_j / \alpha_j$:

$$f = f_0 - t\beta + t x_k + \sum_{i \neq j} (\bar{c}_i - t\alpha_i) x_i.$$

The minimum ratio keeps all displayed coefficients nonnegative. This is dual feasibility in the new basis.

Proof: the minimum ratio covers every remaining column

Let $t = \min_{\alpha_i > 0} \bar{c}_i / \alpha_i \geq 0$. For every remaining nonbasic variable:

$$\alpha_i > 0 \Longrightarrow t \leq \bar{c}_i / \alpha_i \Longrightarrow \bar{c}_i - t\alpha_i \geq 0,$$
$$\alpha_i \leq 0 \Longrightarrow \bar{c}_i - t\alpha_i \geq \bar{c}_i \geq 0.$$

The leaving variable becomes nonbasic with coefficient $t \geq 0$. The entering variable becomes basic with reduced cost zero.

> Thus every dual constraint still holds after the pivot. Feasibility of other primal basic values is not asserted; they must be checked at the next iteration.

Proof: positive ratios give strict dual improvement

$$f_0' = f_0 - t\beta, \qquad \beta < 0, \quad t \geq 0.$$

If $\bar{c}_j > 0$, then $t > 0$, so $f_0' > f_0$. This is an increase of the associated dual objective.

If every pivot has a positive ratio, the objective strictly increases. No basis can repeat, and there are finitely many bases, so the method terminates.

> At termination, either all primal basic values are nonnegative and the pair is optimal, or a negative row has no eligible entering column.

If the ratio is zero, strict improvement fails. This is why a separate anticycling rule is needed in degenerate cases.

Proof: an impossible row gives a certificate and a dual ray

Let ℓ be the selected row. Write $q^\top = e_\ell^\top B^{-1}$, where e_ℓ selects row ℓ. If all tableau entries in that row are nonnegative,

$$q^\top A \geq 0, \qquad q^\top b = \beta < 0.$$

A feasible $x \geq 0$ would give $0 \leq q^\top A x = q^\top b < 0$, a contradiction.

$$p(t) = p - tq: \quad A^\top p(t) \leq c, \qquad b^\top p(t) = b^\top p - t\beta \longrightarrow +\infty.$$

Here $t \geq 0$ and p is the current feasible dual solution. Thus the primal is infeasible and the dual is unbounded.

Lexicographic proof: track full tableau columns

Let $C_j = (\bar{c}_j, u_{1j}, \dots, u_{mj})^\top$, with objective row first and fixed constraint-row positions. Assume every initial C_j is lexicographically positive.

Choose row ℓ, with RHS $\beta < 0$, and let $v_j = u_{\ell j} < 0$ be the entering entry. Since C_j is lex positive but $v_j < 0$, its first nonzero entry occurs **before** row ℓ.

$$\frac{C_j}{|v_j|} \leq_{\text{lex}} \frac{C_i}{|v_i|} \qquad \text{for every eligible } v_i < 0.$$

We prove that all columns remain lex positive and that the full RHS column strictly decreases lexicographically.

Lexicographic proof: columns with nonnegative pivot-row entries

For another column i with $v_i \geq 0$, subtracting the pivot-column multiple gives, outside row ℓ,

$$C_i - \frac{v_i}{v_j} C_j = C_i + \frac{v_i}{|v_j|} C_j.$$

If $v_i > 0$, its first nonzero entry before row ℓ is positive: both columns are lex positive, and C_j already has a positive first nonzero entry before that row.

Changing the pivot-row entry to v_i/v_j cannot change that earlier sign. If $v_i = 0$, the column is unchanged.

> These columns therefore stay lexicographically positive.

Lexicographic proof: eligible columns also stay positive

For another eligible column i, the chosen lexicographic minimum gives

$$D_i = C_i - \frac{|v_i|}{|v_j|} C_j \ \geq_{\text{lex}} \ 0.$$

Its entry in row ℓ is zero. The actual pivot replaces that entry by $v_i / v_j > 0$.

If the first nonzero entry of this difference is before the pivot row, it is positive. Otherwise, the positive replacement in the pivot row becomes its first nonzero entry.

> The entering column itself becomes e_ℓ, also lex positive. This proves preservation for every column.

Lexicographic proof: no ordered basis can repeat

Let $C_0 = (-f_0, \bar{b}_1, \dots, \bar{b}_m)^\top$ be the full RHS column. Outside the pivot row,

$$C_0' = C_0 - \frac{\beta}{v_j} C_j, \qquad \frac{\beta}{v_j} > 0.$$

The first nonzero entry of C_j is positive and occurs before row ℓ. Hence the first changed entry of C_0 decreases. Normalizing the later pivot-row entry cannot undo this.

$$C_0' <_{\text{lex}} C_0.$$

Every ordered basis determines one tableau, and there are finitely many ordered bases. Strict lexicographic decrease rules out repetition. Therefore the method terminates.

Example 4.8 in tableau form: initial state

Primal · Example 4.8

$$\begin{aligned} \min \quad & x_1 + x_2 \\ & x_1 + 2x_2 \geq 2, \quad x_1 \geq 1, \\ & x_1, x_2 \geq 0. \end{aligned}$$

Dual

$$\begin{aligned} \max \quad & 2p_1 + p_2 \\ & p_1 + p_2 \leq 1, \quad 2p_1 \leq 1, \\ & p_1, p_2 \geq 0. \end{aligned}$$

Row	Column 0	x_1	x_2	x_3	x_4
R_0	0	1	1	0	0
x_3	-2	-1	-2	1	0
x_4	-1	-1	0	0	1

Column 0 is the RHS in constraint rows and $-f_0$ in R_0, where $f = f_0 + \sum_{j \in \mathcal{N}} \bar{c}_j x_j$. Columns always follow natural variable order.

Both $x_3 = -2$ and $x_4 = -1$ are negative. Our smallest-index convention selects x_3 to leave; ratios 1 and $1/2$ select x_2 to enter.

Example 4.8 in tableau form: after pivot 1

Primal · Example 4.8

$$\begin{aligned} \min \quad & x_1 + x_2 \\ & x_1 + 2x_2 \geq 2, \quad x_1 \geq 1, \\ & x_1, x_2 \geq 0. \end{aligned}$$

Dual

$$\begin{aligned} \max \quad & 2p_1 + p_2 \\ & p_1 + p_2 \leq 1, \quad 2p_1 \leq 1, \\ & p_1, p_2 \geq 0. \end{aligned}$$

Row	Column 0	x_1	x_2	x_3	x_4
R_0	-1	$1/2$	0	$1/2$	0
x_2	1	$1/2$	1	$-1/2$	0
x_4	-1	-1	0	0	1

Column 0 gives $x_2 = 1$, $x_4 = -1$, and objective $f_0 = 1$. The negative x_4 row must leave; only x_1 is eligible to enter.

The negative coefficient -1 in this equality row corresponds to the positive coefficient in the dictionary $x_4 = -1 + x_1$.

Example 4.8 in tableau form: after pivot 2

Primal · Example 4.8

$$\begin{aligned} \min \quad & x_1 + x_2 \\ & x_1 + 2x_2 \geq 2, \quad x_1 \geq 1, \\ & x_1, x_2 \geq 0. \end{aligned}$$

Dual

$$\begin{aligned} \max \quad & 2p_1 + p_2 \\ & p_1 + p_2 \leq 1, \quad 2p_1 \leq 1, \\ & p_1, p_2 \geq 0. \end{aligned}$$

Row	Column 0	x_1	x_2	x_3	x_4
R_0	$-3/2$	0	0	$1/2$	$1/2$
x_1	1	1	0	0	-1
x_2	$1/2$	0	1	$-1/2$	$1/2$

Basic quantities $(1, 1/2)$ and all reduced costs are nonnegative. The objective value is $f^* = 3/2$.

Rows are reordered only for this final display. A lexicographic iteration keeps their positions fixed.

Why each basis gives a basic dual solution

A basis consists of m independent columns $A_{B(1)}, \dots, A_{B(m)}$. The associated dual solution satisfies

$$A_{B(i)}^{\top} p = c_{B(i)}, \qquad i = 1, \dots, m.$$

These are m independent active equalities in the m-dimensional variable p. Their unique intersection is therefore a basic dual solution.

> It is feasible exactly when all remaining inequalities $A_j^{\top} p \le c_j$ also hold—equivalently, when all reduced costs are nonnegative.

A pivot changes one basic column, so consecutive feasible dual bases share m−1 independent active equalities. Their points are adjacent or coincide.

Example 4.9: all six bases

Primal · Example 4.9

$$\begin{aligned} \min \quad & 3x_1 + x_2 \\ & x_1 + x_2 \geq 2, \quad 2x_1 - x_2 \geq 0, \\ & x_1, x_2 \geq 0. \end{aligned}$$

Dual

$$\begin{aligned} \max \quad & 2p_1 \\ & p_1 + 2p_2 \leq 3, \quad p_1 - p_2 \leq 1, \\ & p_1, p_2 \geq 0. \end{aligned}$$

Basic variables	Primal (x_1,x_2)	Dual (p_1,p_2)
(x_3, x_4)	$(0, 0)$	$(0, 0)$
(x_1, x_3)	$(0, 0)$	$(0, 3/2)$
(x_2, x_3)	$(0, 0)$	$(0, -1)$
(x_2, x_4)	$(0, 2)$	$(1, 0)$
(x_1, x_2)	$(2/3, 4/3)$	$(5/3, 2/3)$
(x_1, x_4)	$(2, 0)$	$(3, 0)$

Every row comes from setting nonbasic primal variables to zero and setting basic-column dual constraints active. Distinct bases need not mean distinct primal points.

Why a zero nonbasic reduced cost means dual degeneracy

At the dual point associated with B, the m basic-column constraints are active. For any other column,

$$\bar{c}_j = c_j - A_j^\top p = 0 \quad \Longleftrightarrow \quad A_j^\top p = c_j.$$

Thus a zero nonbasic reduced cost supplies an additional active constraint beyond the m constraints defining the basis.

> This can produce multiple dual bases at one dual point. Conversely, every additional active nonbasic-column constraint has reduced cost zero.

This observation concerns dual degeneracy; a zero primal basic value is a separate condition.

Which property survived the model change?

Change from an optimal basis	Property preserved	Natural next step
Only RHS b changes	Dual feasibility	Check basic values; use dual simplex if needed.
Only costs c change	Primal feasibility	Check reduced costs; use primal simplex if needed.
A new nonnegative variable is added	Old primal point extends with zero	Check the new reduced cost.
A new ≤ constraint is added	Old dual solution extends with zero	Check the new slack; dual simplex if negative.

This is a starting-state diagnosis, not a guarantee of fewer pivots. Sensitivity analysis will calculate the ranges where no pivot is needed.

The initial tableau and the same three row operations

Primal · Example 4.7

$$\begin{aligned}\min\quad & 2x_1 + 6x_2 + 10x_3 \\ & -2x_1 + 4x_2 + x_3 + x_4 = 2, \\ & 4x_1 - 2x_2 - 3x_3 + x_5 = -1, \\ & x_1, x_2, x_3, x_4, x_5 \geq 0.\end{aligned}$$

Dual

$$\begin{aligned}\max\quad & 2p_1 - p_2 \\ & -2p_1 + 4p_2 \leq 2, \\ & 4p_1 - 2p_2 \leq 6, \\ & p_1 - 3p_2 \leq 10, \\ & p_1 \leq 0, \quad p_2 \leq 0.\end{aligned}$$

Row	Column 0	x_1	x_2	x_3	x_4	x_5
R_0	0	2	6	10	0	0
x_4	2	−2	4	1	1	0
x_5	−1	4	−2	−3	0	1

Let R_1 be the x_4 row and R_2 the x_5 row. Normalize, then eliminate:

$$R_2' = -\tfrac{1}{2}R_2, \qquad R_1' = R_1 - 4R_2', \qquad R_0' = R_0 - 6R_2'.$$

The upper-left entry becomes -3. Compare the completed tableau, with objective 3.

The smallest ratio preserves every dual inequality

After solving the selected row for x_j, its reduced cost is eliminated. Let $t = \bar{c}_j/\alpha_j \geq 0$.

$$\text{New coefficient of } x_i = \bar{c}_i - t\alpha_i, \qquad \text{new coefficient of } x_k = t.$$

Other coefficient	Why it stays nonnegative
$\alpha_i > 0$	$t \leq \bar{c}_i/\alpha_i$ by the minimum ratio.
$\alpha_i \leq 0$	$\bar{c}_i - t\alpha_i \geq \bar{c}_i \geq 0$.

This is why taking any variable with a helpful sign is not enough. The objective row determines how far the dual variables may change.

The same basis explains both plotted points

Primal · Example 4.8

$$\begin{aligned} \min \quad & x_1 + x_2 \\ & x_1 + 2x_2 \geq 2, \quad x_1 \geq 1, \\ & x_1, x_2 \geq 0. \end{aligned}$$

Dual

$$\begin{aligned} \max \quad & 2p_1 + p_2 \\ & p_1 + p_2 \leq 1, \quad 2p_1 \leq 1, \\ & p_1, p_2 \geq 0. \end{aligned}$$

Label	Basic variables	Primal point (x_1,x_2)	Dual point (p_1,p_2)
A	(x_3, x_4)	$(0, 0)$	$(0, 0)$
B	(x_2, x_4)	$(0, 1)$	$(1/2, 0)$
C	(x_1, x_2)	$(1, 1/2)$	$(1/2, 1/2)$
D	(x_1, x_3)	$(1, 0)$	$(0, 1)$
E	(x_1, x_4)	$(2, 0)$	$(1, 0)$

The algorithm used A → B → C. E is dual infeasible and D is primal infeasible. At C, both tests pass.

Adjacent dual bases share one active constraint here. In higher dimensions, consecutive bases share all but one of their basic-column equalities.

Recall the optimal dual variables

Primal · Example 4.6

$$\begin{aligned} \min \quad & 13x_1 + 10x_2 + 6x_3 \\ & 5x_1 + x_2 + 3x_3 = 8, \\ & 3x_1 + x_2 = 3, \qquad x \geq 0. \end{aligned}$$

Dual

$$\begin{aligned} \max \quad & 8p_1 + 3p_2 \\ & 5p_1 + 3p_2 \leq 13, \\ & p_1 + p_2 \leq 10, \quad 3p_1 \leq 6, \\ & p_1, p_2 \in \mathbb{R}. \end{aligned}$$

p_1 corresponds to the first equality and p_2 to the second.

$$p^* = (2, 1), \qquad 8(2) + 3(1) = 19.$$

The identity $13x_1 + 10x_2 + 6x_3 = 2(5x_1 + x_2 + 3x_3) + (3x_1 + x_2) + 7x_2$ proves the matching bound.

Used ingredients are priced exactly at their cost

Primal · Example 4.6

$$\begin{aligned} \min \quad & 13x_1 + 10x_2 + 6x_3 \\ & 5x_1 + x_2 + 3x_3 = 8, \\ & 3x_1 + x_2 = 3, \qquad x \geq 0. \end{aligned}$$

Dual

$$\begin{aligned} \max \quad & 8p_1 + 3p_2 \\ & 5p_1 + 3p_2 \leq 13, \\ & p_1 + p_2 \leq 10, \quad 3p_1 \leq 6, \\ & p_1, p_2 \in \mathbb{R}. \end{aligned}$$

$$x^* = (1, 0, 1), \qquad c - A^\top p^* = (0, 7, 0), \qquad p^* = (2, 1).$$

$$f = 19 + 7x_2.$$

Ingredient 2 adds cost after the basic ingredients are adjusted, so it is unused. Used ingredients 1 and 3 have zero gaps.

A zero gap permits positive use; it does not force positive use.

Complete the tied-ratio example's pivot

Primal · variant: cost of x_3 is 9

$$\begin{aligned}\min \quad & 2x_1 + 6x_2 + 9x_3 \\ & -2x_1 + 4x_2 + x_3 + x_4 = 2, \\ & 4x_1 - 2x_2 - 3x_3 + x_5 = -1, \\ & x_1, x_2, x_3, x_4, x_5 \geq 0.\end{aligned}$$

Dual

$$\begin{aligned}\max \quad & 2p_1 - p_2 \\ & -2p_1 + 4p_2 \leq 2, \\ & 4p_1 - 2p_2 \leq 6, \\ & p_1 - 3p_2 \leq 9, \\ & p_1 \leq 0, \quad p_2 \leq 0.\end{aligned}$$

$$\begin{aligned}x_3 &= \tfrac{1}{3} + \tfrac{4}{3}x_1 - \tfrac{2}{3}x_2 + \tfrac{1}{3}x_5, \\ x_4 &= \tfrac{5}{3} + \tfrac{2}{3}x_1 - \tfrac{10}{3}x_2 - \tfrac{1}{3}x_5, \\ f &= 3 + 14x_1 + 0x_2 + 3x_5.\end{aligned}$$

$x = (0, 0, 1/3, 5/3, 0)$ and $p = (0, -3)$ are feasible, with value 3. The zero nonbasic reduced cost belongs to x_2.